

Workforce Attrition Analysis

Executive Business Report

Company: NovaTech Solutions

Department: Human Resources

Analysis Type: Workforce attrition and retention analysis

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Data Source: Synthetic HR employee dataset

Tool: Microsoft Excel

Executive Summary

This analysis examined workforce attrition across 1,468 employees to identify the employee groups and workplace conditions most strongly associated with turnover.

A total of 226 employees left the organization, producing an overall attrition rate of 15.4%. The analysis indicates that attrition is concentrated among employees who work overtime, early-career employees, recently hired employees, Sales staff, and employees reporting low job satisfaction. Overtime represents the strongest observed association with attrition in the dataset. Employees who worked overtime had a 23.6% attrition rate, compared with 12.3% among employees who did not work overtime.

Sales also requires management attention. It recorded the highest departmental attrition rate at 19.6%, representing 87 departures among 443 employees. Within job roles, Sales Representatives had a 21.7% attrition rate and accounted for 33 departures, making the role a meaningful retention priority.

Attrition was also strongly concentrated among younger and recently hired employees. Employees under age 25 had a 26.3% attrition rate, while employees with less than one year of tenure had a 21.3% attrition rate. These findings indicate a potential onboarding and early-career retention gap.

Key Findings

Overall workforce attrition

The organization's overall attrition rate was 15.4%, with 226 departures among 1,468 employees. This means approximately one in every 6.5 employees left the organization.

Departmental risk

Sales had the highest departmental attrition rate at 19.6%, compared with 17.5% in Human Resources and 13.3% in Research and Development.

Because Sales also has a substantial employee population, its attrition represents both a high rate and a meaningful volume of departures.

Job-role risk

HR Business Partner recorded the highest observed job-role attrition rate at 23.8%. However, this result is based on only 21 employees and five departures, so it should be treated as a high-rate, low-volume segment requiring further investigation rather than as the company's primary attrition driver. Sales Representative presented a more operationally significant concern, with a 21.7% attrition rate, 33 departures, and a population of 152 employees.

Overtime

Employees who worked overtime had an attrition rate of 23.6%, compared with 12.3% among employees who did not work overtime. This was the strongest observed association with attrition among the variables analyzed.

The result does not establish that overtime directly causes attrition, but it provides a clear area for management investigation.

Age and early-career retention

Employees under age 25 had the highest age-group attrition rate at 26.3%. This was more than double the 11.8% rate among employees aged 55 and above.

The pattern indicates that younger and early-career employees may require greater career guidance, development opportunities, managerial support, and role clarity.

Employee tenure

Employees with less than one year of tenure had the highest tenure-based attrition rate at 21.3%. The rate fell to 10.8% among employees with ten or more years of service.

This represents the largest decline across the tenure categories and suggests a possible weakness in recruitment fit, onboarding, first-year engagement, or early career development.

Job satisfaction

Employees reporting low job satisfaction had an attrition rate of 20.2%, compared with 12.9% among employees reporting very high satisfaction.

The pattern was generally downward as satisfaction increased, indicating that employee experience and engagement should be central to retention planning.

Income

The Low Income category recorded a 29.4% attrition rate. However, the category contained only 17 employees, making the finding highly sensitive to a small number of departures.

The more reliable comparison is between Middle Income employees, with an 18.2% attrition rate, and High Income employees, with a 14.8% rate. Income should therefore be monitored as a potential factor rather than presented as a confirmed primary driver.

Demographic factors

Gender and marital status showed relatively limited variation in attrition. Rates generally remained between approximately 14.2% and 16.5%.

These variables should not be treated as major retention drivers based on the current descriptive analysis.

Recommendations

1. Review overtime and workload allocation

HR and department leaders should examine overtime frequency, workload distribution, staffing levels, scheduling, and manager expectations, particularly in Sales and other overtime-heavy teams. A useful first step would be to compare overtime hours, workload, absence, satisfaction, and turnover at the team and manager levels.

2. Strengthen first-year onboarding

Introduce a structured first-year employee retention program that includes:

- Preboarding communication
- Role and expectation clarification
- 30-, 60-, and 90-day manager check-ins
- Six- and twelve-month career conversations
- Buddy or mentoring support
- Early employee engagement surveys

3. Investigate Sales Representative working conditions

Conduct targeted analysis and employee interviews for the Sales Representative population. Areas to investigate include:

- Sales targets and workload
- Commission and incentive structure
- Managerial support
- Territory allocation
- Career progression

- Training quality
- Schedule flexibility
- Overtime requirements

The HR Business Partner group should also be reviewed, but its 23.8% rate should be interpreted cautiously because of the small population.

4. Connect satisfaction data to management action

Managers should receive regular summaries of employee satisfaction and engagement results. Low scores should trigger structured follow-up conversations, career-development planning, and documented improvement actions.

5. Monitor compensation as a hypothesis

The high attrition rate in the Low Income group should be monitored but not treated as conclusive evidence because of its small sample size.

Additional analysis should compare salary position within role, job level, performance, tenure, and market benchmarks.

6. Establish recurring attrition reporting

The analysis should be repeated monthly or quarterly when connected to live HR data. The dashboard should track:

- Overall attrition
- Voluntary and involuntary departures
- First-year attrition
- Overtime-related attrition
- Department and manager risk
- Critical-role attrition
- Satisfaction trends
- Retention action progress

Limitations

This analysis is descriptive and identifies associations, not causal relationships.

Additional limitations include:

- The dataset is synthetic.
- Nineteen employees have no recorded department and are grouped as Unknown Department.
- Some categories, including HR Business Partner and Low Income, have small sample sizes.
- No exit-interview, manager, promotion-history, market-pay, or employee-engagement data was available.
- The analysis represents an employee snapshot rather than a complete longitudinal workforce history.

Conclusion

The analysis identifies overtime, Sales roles, early tenure, younger age groups, and low job satisfaction as the most important areas for further retention investigation.

Management should prioritize workload review and first-year retention initiatives while continuing to collect more detailed employee-experience and exit data. A recurring monitoring process would allow HR to assess whether these interventions reduce attrition over time.



Overtime

Overtime	Attrition Rate
No	0.12
Yes	0.24

Age Group

Age Group	Attrition Rate
25-34	0.16
35-44	0.13
45-54	0.15
55+	0.12
Under 25	0.27

Tenure

Tenure	Attrition Rate
0-1 Years	0.22
10+ Years	0.11
2-4 Years	0.12
5-9 Years	0.13

Job Satisfaction

Job Satisfaction	Attrition Rate
High	0.13
Low	0.20
Medium	0.15
Very High	0.13

SEGMENT RISK MATRIX

ANALYST TAKEAWAYS

Critical | HR Business Partner: 23.8% attrition (5 exits / 21 employees).

Scale | R&D: 126 exits, 56% of all departures.

Age | Under 25: 26.3% attrition; base of 133 employees.

Data quality | Unknown gender: 11.7% attrition across 60 employees.

Overtime | No: 12.3% vs Yes: 23.6% attrition.

Interpretation note: distinguish high-rate small segments from high-volume drivers before prioritizing interventions.